EM Continued & Probabilistic PCA

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If you cannot answer a hard question directly,
reframe it into an easier question you can (hopefully) answer exactly

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Recall: EM for GMM

The Expectation Maximization (EM) algorithm is a powerful technique for finding maximum likelihood solutions for latent variable models

E Step

- Solve the inference problem: *which Gaussian generated each datapoint?* or more precisely, *how responsible is each Gaussian for generating each datapoint?*
- Compute the posterior probabilities, $p(z_k = 1|\mathbf{x})$

M Step

- Use the assigned *responsibilities* (the posterior probabilities from the E step) to re-estimate the means, covariances, and mixing coefficients

ELBO: Toward a general perspective on the EM

Consider a probabilistic model with all observed variables $\mathbf{X}$ and latent variables $\mathbf{Z}$. Introducing a distribution $q(\mathbf{Z})$, we can write:

$$\ln p(\mathbf{X}|\boldsymbol{\theta}) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$$

(assuming discrete $\mathbf{Z}$, derivation is identical for continuous $\mathbf{Z}$ with integration replacing summation)

$$= \ln \sum_{\mathbf{Z}} \frac{q(\mathbf{Z})}{q(\mathbf{Z})} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$$

$$= \ln \mathbb{E}_q \left[\frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right]$$

$$\geq \mathbb{E}_q \left[\ln \frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right]$$

… Jensen's Inequality

$$= \mathcal{L}(q, \boldsymbol{\theta})$$

… Evidence Lower Bound, or ELBO

E and M Step Details: Blackboard

Continuous LVMs: Probabilistic PCA

Let's revisit the Gaussian model

We assume we have a dataset $\mathbf{X}$ with observations $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \in \mathbb{R}^D$

The multivariate Gaussian has two parameters: $\boldsymbol{\mu} \in \mathbb{R}^D, \quad \boldsymbol{\Sigma} \in \mathbb{S}_+^D$

MLE yields

$$\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{N} \mathbf{x}_i, \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{n}(\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^T$$

$$\mathbb{S}_+^D = \{A \in \mathbb{R}^{n \times n} : A = A^T \text{ and } x^T A x > 0 \text{ for all } x \in \mathbb{R}^n \text{ such that } x \neq 0\}$$

What happens in high dimensions?

Now consider $D \gg N$, what can go wrong?

$\hat{\Sigma}$ is a $D \times D$ matrix, with rank $\min(N, D)$ -> noninvertible 😖

Aside: While the covariance matrix contains many parameters (e.g. ~300K for MNIST dimensional data), deep generative models often contain an order of magnitude more parameters. One of the factors driving deep learning's success is how well the techniques scale.

Adapted from Pierre-Alexandre Mattei's introduction slides to PPCA

So, what can we do?

A simple idea: replace the full covariance with something more parsimonious[1]!

Examples :

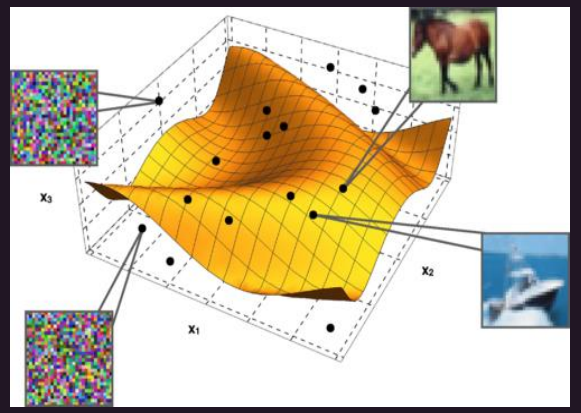
- Spherical $\Sigma = \lambda I$
- Diagonal $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_D^2)$
- Low rank + isotropic noise $\Sigma = \mathbf{W}\mathbf{W}^T + \sigma^2 \boldsymbol{I}$ (PPCA structure)

[1] The law of parsimony, or Occam's razor — preference for the simplest 'explanation' (here, the number of parameters) that still fits the data

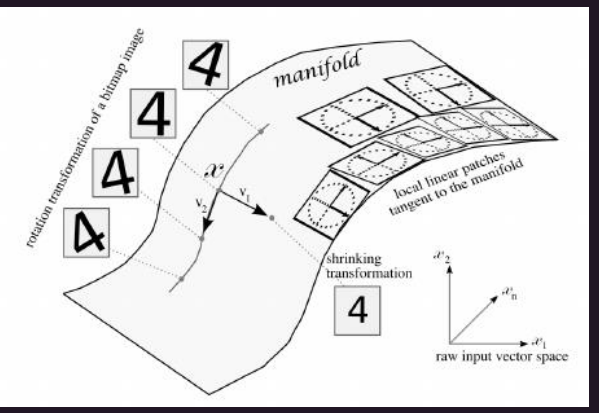
Adapted from Pierre-Alexandre Mattei's introduction slides to PPCA

The Manifold Hypothesis

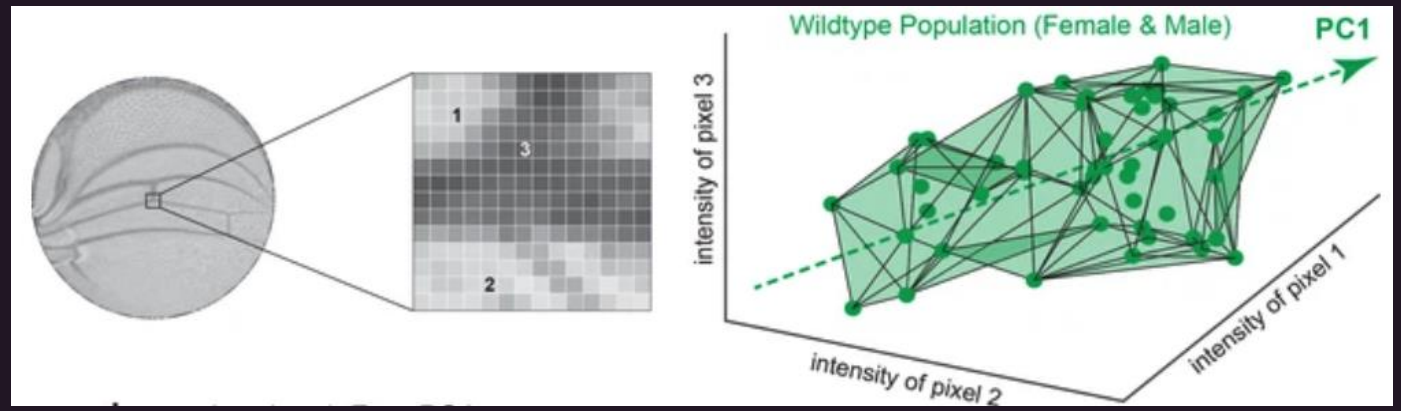
Most naturally occurring high-dimensional data lies on a low-dimensional manifold embedded in that high-dimensional space



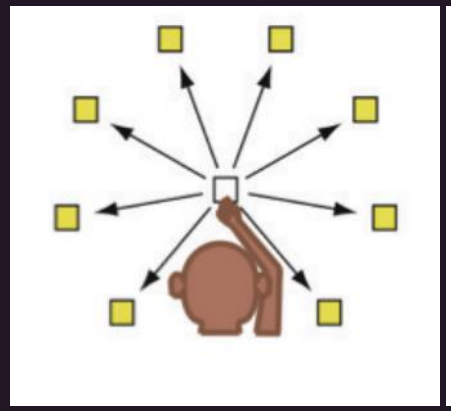
CIFAR10 examples. Goldt et al.



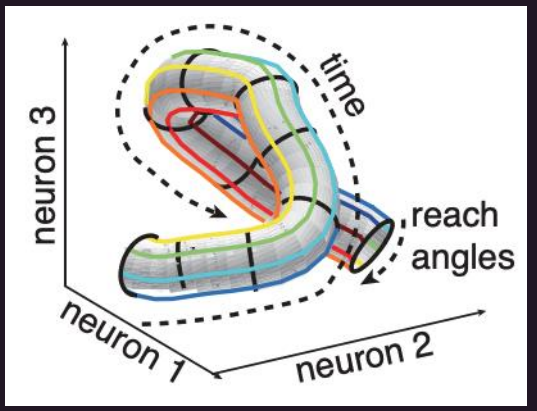
Digit images. Bengio



Biology - Wing structure variation. Alba et al.



Neural activity. Gao et al.



"Testing the manifold hypothesis". C. Fefferman, S. Mitter, and H. Narayanan. 2016

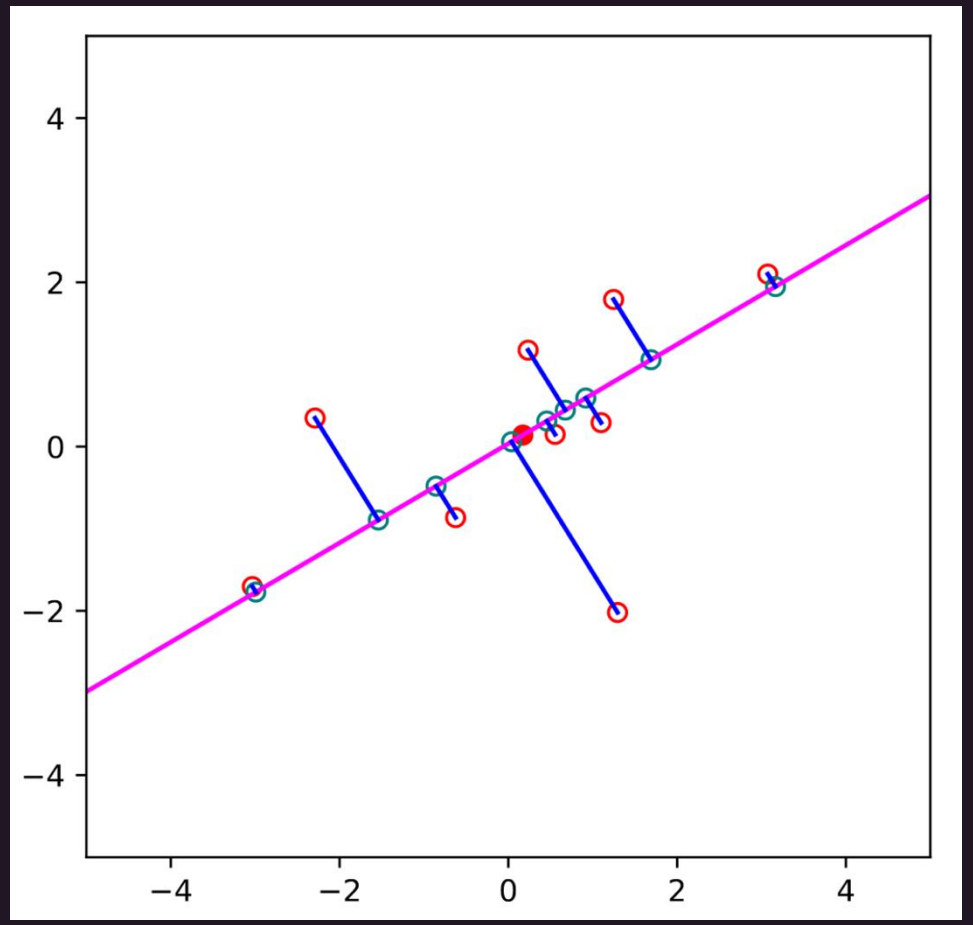
"Low-dimensional manifolds in neuroscience and evolution". Joram Keijser. 2022

Principle Component Analysis (PCA)

A linear approximation of the manifold hypothesis.

Like saying *“Sure, the data is on a manifold — and I’m going to assume it’s flat.”*

See reading material for the minimum error and maximum variance formulations of PCA



Recall: The Latent Variable Model

A latent variable model defines a probability distribution

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$$

We assume we have a dataset $\mathbf{X}$ with iid observations $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \in \mathbb{R}^D$

We now also now assume associated iid latents $\{\mathbf{z}_1, \ldots, \mathbf{z}_N\} \in \mathbb{R}^Q, Q < D$

What choices can we make to formulate a probabilistic version of PCA?

Probabilistic PCA

Simple choices:

- The latent $p(\mathbf{z})$ is a zero-mean unit-covariance Gaussian
- The conditional $p(\mathbf{x}|\mathbf{z})$ is also Gaussian where the mean is a linear function of $\mathbf{z}$ and the covariance is isotropic

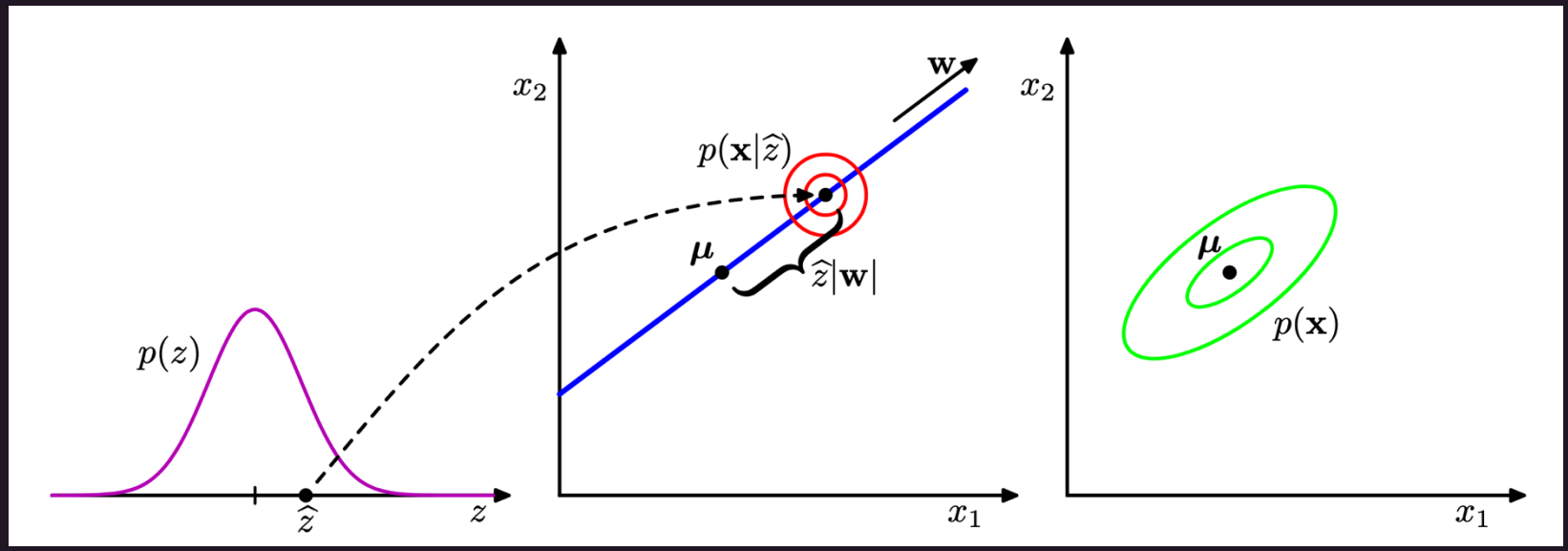
Can you write down the probabilistic PCA model?

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I})$$

$$p(\mathbf{x}|\mathbf{z}, \mathbf{W}, \boldsymbol{\mu}, \sigma) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$$

PPCA: Generative Perspective

$$\mathbf{x} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}$$



What's the dimension of W?

Note that this perspective maps from latent space to data space! Compare this to the conventional PCA view!

PPCA: Likelihood

$$p(\mathbf{x}) = \int p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

$$\begin{aligned}
\mathbb{E}[\mathbf{x}] &= \mathbb{E}[\mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}] = \boldsymbol{\mu} \\
\mathrm{cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^{\mathrm{T}}] \\
&= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^{\mathrm{T}}\mathbf{W}^{\mathrm{T}}] + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^{\mathrm{T}}] \\
&= \mathbf{W}\mathbf{W}^{\mathrm{T}} + \sigma^2\mathbf{I}
\end{aligned}$$

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{C}), \quad \mathbf{C} \in \mathbb{S}_+^D = \mathbf{W}\mathbf{W}^{\mathrm{T}} + \sigma^2\mathbf{I}$$

PPCA: Likelihood

The interpretation:

- $\mathbf{W}\mathbf{W}^{\mathrm{T}}$: low-rank ($\leq Q$) structure from the latent space
- $\sigma^2 \boldsymbol{I}$: isotropic noise ≤
- PPCA models data lying near a linear subspace plus some Gaussian noise

The MLE estimate of $\mathbf{W}$ can be computed in closed-form, and is directly related to PCA! The MLE can also be computed by SVD of the covariance (see Reading Material)

The story so far

- We have seen latent variable models with both discrete and continuous latent variables
- We have also seen how to formulate training objectives for latent variable models

Next week: Deep Latent Variable Models (DLVMs)! Think of DLVMs as non-linear PPCA, or as a mixture model with continuous $\mathbf{z}$